HW#5

Q1) h, x_t, h_t, u_t (Δh_t from $x_{t-1} - x_t$)

1) $x_t = A x_{t-1} + B u_t + \varepsilon_t$

$z_t = C x_{t-1} + \delta_t$

$h_t = [1]\, h_{t-1} + [1]\, \Delta h_t + N(0, 0.25)$

$z_t = [1]\, h_{t-1} + N(0, 0.1)$

$\varepsilon_t \sim N(0, \sigma_u^2 = 0.25)$

$\delta_t \sim N(0, \sigma_z^2 = 0.1)$

$x_t = h_t \quad A = 1$

$B = 1$

$u_t = \Delta h_t \quad \delta_t =$

$C = 1$

2) Estimate the state after

$u_1 = 0.5,\ z_1 = 3.6,\ u_2 = 0.25,\ z_2 = 3.7$

Find $\bar{\mu}_t, \bar{\sigma}_t^2, M_t, \sigma_t^2 + K_t$ for each timestep

$bel(x_0) \sim N(\mu_0 = 3, \sigma_0^2 = 0.5)$

~~$\bar{\mu}_t = A_t + B_t u_t$~~

$\bar{\mu}_1 = A\mu_0 + B u_0$

$= [1](3) + [1]\, 0.5$

$\bar{\mu}_1 = 3.5$

$\bar{\sigma}_1^2 = A \sigma_0^2 A^T + R_t$

$\bar{\sigma}_1^2 = [1]\, 0.5\, [1] + 0.25$

$\bar{\sigma}_1^2 = 0.75$

$K_1 = \bar{\sigma}_1^2 C^T (C_1 \bar{\sigma}_1^2 C_1^T + Q_t)^{-1}$

$= (0.75)[1]([1](0.75)[1] + 0.1)^{-1}$

$= 0.75 \left(\frac{1}{0.85}\right)$

$K_1 = 0.882$

$\mu_1 = \bar{\mu}_1 + K_1 (z_1 - [1]\bar{\mu}_1)$

$= 3.5 + 0.882(3.6 - 3.5)$

$\mu_1 = 3.589$

$\sigma_1^2 = \bar{\sigma}_1^2 - K_1 C_1 \bar{\sigma}_1^2$

$\sigma = 0.75 - 0.882[1](0.75)$

$\sigma_1^2 = 0.0862$

$\bar{\mu}_2 = A\mu_1 + B u_2$

$\bar{\mu}_2 = 3.589 + 0.25$

$\bar{\mu}_2 = 3.839$

$\bar{\sigma}_2^2 = A\sigma_1^2 A^T + R_t$

$= (0.0382) + 0.25$

$\bar{\sigma}_2^2 = 0.3382$

$K_2 = \bar{\sigma}_2^2 C^T (C \bar{\sigma}_2^2 C^T + Q_t)^{-1}$

$= (0.3382)(0.3382 + 0.1)^{-1}$

$K_2 = 0.7718$

$\mu_2 = \bar{\mu}_2 + K_2(z_2 - C_2 \bar{\mu}_2)$

$= 3.839 + (0.7718)(3.7 - 3.839)$

$\mu_2 = 3.732$

$\sigma_2^2 = \bar{\sigma}_2^2 - K_2 C_2 \bar{\sigma}_2^2$

$= 0.3382 - (0.7718)(0.3382)$

$\sigma_2^2 = 0.0772$

Q2) $bel(x_t) = p(x_t \mid u_{1:t}, z_{1:t})$

$y = Ax + b \longrightarrow \mu_y = A\mu_x + b \quad \Sigma_y = A\Sigma_x A^T$

prediction

$\xi = \begin{bmatrix} x_t \\ x_{t-1} \end{bmatrix} = F x_{t-1} + G u_t + H \varepsilon_t$

$F = \begin{bmatrix} A \\ I \end{bmatrix} \quad G = \begin{bmatrix} B \\ 0 \end{bmatrix} \quad H = \begin{bmatrix} I \\ 0 \end{bmatrix}$

$\mu_\xi = \begin{bmatrix} A\mu_{t-1} + B u_t \\ \mu_{t-1} \end{bmatrix} \quad \Sigma_\xi = F\Sigma_{t-1}F^T + HRH^T = \begin{bmatrix} A\Sigma_{t-1}A^T + R & A\Sigma_{t-1} \\ \Sigma_{t-1}A^T & \Sigma_{t-1} \end{bmatrix}$

$\bar{\mu}_t = A\mu_{t-1} + B u_t$

$\bar{\Sigma}_t = A\Sigma_{t-1}A^T + R$

correction

$\Sigma = \begin{bmatrix} x_t \\ z_t \end{bmatrix} = Dx_t + E\delta_t$

$D = \begin{bmatrix} I \\ C \end{bmatrix} \quad E = \begin{bmatrix} 0 \\ I \end{bmatrix}$

$\mu_\Sigma = \begin{bmatrix} \bar{\mu}_t \\ C\bar{\mu}_t \end{bmatrix} \quad \Sigma_\Sigma = D\bar{\Sigma}_t D^T + EQE^T = \begin{bmatrix} \bar{\Sigma}_t & \bar{\Sigma}_t C^T \\ C\bar{\Sigma}_t & C\bar{\Sigma}_t C^T + Q \end{bmatrix}$

$\mu_t = \bar{\mu}_t + \bar{\Sigma}_t C^T (C\bar{\Sigma}_t C^T + Q)^{-1}(z_t - C\bar{\mu}_t)$

$\Sigma_t = \bar{\Sigma}_t - \bar{\Sigma} C^T (C\bar{\Sigma}_t C^T + Q)^{-1} C\bar{\Sigma}_t$

$K_t = \bar{\Sigma} C^T (C\bar{\Sigma}_t C^T + Q)^{-1} = \Sigma_{\alpha\beta}\Sigma_{\beta\beta}^{-1}$

$\mu' = \mu_\alpha + \Sigma_{\alpha\beta}\Sigma_{\beta\beta}^{-1}(\beta - \mu_\beta)$

$\Sigma' = \Sigma_{\alpha\alpha} - \Sigma_{\alpha\beta}\Sigma_{\beta\beta}^{-1}\Sigma_{\beta\alpha}$

$\mu_t = \bar{\mu}_t + K_t(z_t - C\bar{\mu}_t)$

$\Sigma_t = \bar{\Sigma}_t - K_t C\bar{\Sigma}_t = (I - K_t C)\bar{\Sigma}_t$